

Assessment on Separating Hydrofluoroolefins from Hydrofluorocarbons at the Azeotropic Mixture R513A by Using Fluorinated Ionic Liquids: A Soft-SAFT Study

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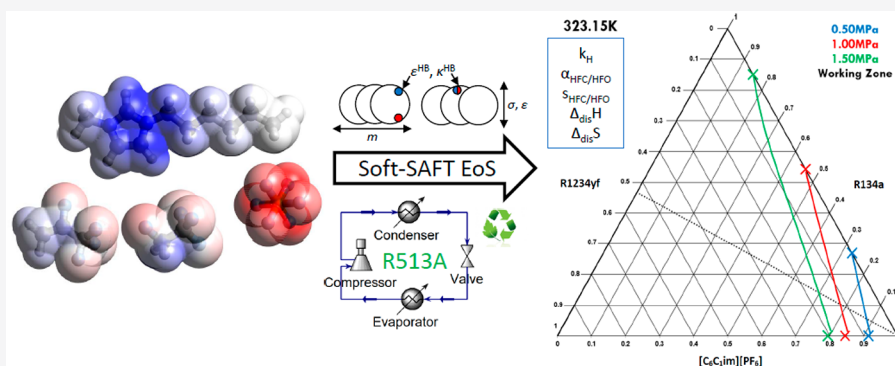
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ABSTRACT: In recent years, the commercial blend R513A (integrated by R134a and R1234yf) has been proposed to replace R134a in air conditioning applications, given the lower global warming potential of R513A versus R134a. However, little is known about the best approach to recover these compounds by adequately separating them after the end of life of the equipment. In this contribution, we present a computational study of the capability of three fluorinated ionic liquids (imidazolium-based ILs with the $[\text{BF}_4]^-$, $[\text{PF}_6]^-$, and $[\text{TF}_2\text{N}]^-$ anions) to selectively separate R134a and R1234yf from the R513A mixture. A thermodynamic analysis of the absorption of these gases in the selected ionic liquids has been performed with the soft-SAFT molecular-based equation of state (EoS), with the model validated using available experimental data. Soft-SAFT has then been used to predict the Henry constant, the solvation enthalpy and entropy, and the ideal selectivity in order to assess the best solvent to carry out the separation. Results indicate that $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ exhibits the highest ideal selectivity and the highest difference of molar enthalpy between both compounds. Hence, the ternary diagrams with this IL, R134a, and R1234yf are predicted, showing the competitive selectivity between both gases at the R513A mixture composition and evaluating the effect of pressure and temperature in the system. It has been observed that the competitive selectivity is similar to the ideal selectivity at the studied conditions. From the results obtained, it is recommended to work at low temperature and moderate pressure conditions to carry out this separation with $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$.

1. INTRODUCTION

It is well-known that one of the current society challenges comes from the necessity to urgently decrease the emissions of gases that contribute to global warming. While the major focus is related to carbon dioxide (CO_2), the contributions coming from fluorinated gases used in the refrigeration industry, mainly hydrofluorocarbons (HFCs), also have a severe environmental impact, due to their elevated global warming potential (GWP). In the search for alternative refrigerants, the use of hydrofluoroolefins (HFOs) represents an attractive option due to their very low GWPs. HFOs are part of the largest group of compounds identified as new candidates with a collection of adequate thermophysical properties to become the refrigerants of the future.^{1,2} However, their use as single compounds is restricted by a combination of cost production

and nonzero flammability (the most common R1234yf and R1234ze(E) HFOs fall in the A2L—mildly flammable—category according to the ASHRAE classification,³ being restricted in systems located indoors with large charge sizes),⁴ and also because in some cases the required thermophysical properties do not match the required values for the specific application. In this regard, the only possible solution is the study of mixtures of compounds that are able to

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provide a compromise between good performance (adequate thermophysical properties to carry out their final purpose), safety (relatively low flammability), and low environmental impact (reduced GWP).¹

Currently, several new blends based on a combination of HFCs and HFOs have entered the market as fourth generation refrigerants. One of the most common is named R513A, a commercial mixture of the HFC R134a (44 wt %) and the HFO R1234yf (56 wt %), whose properties are adequate for air conditioning applications. R513A is commonly denominated as azeotropic because the R134a + R1234yf system exhibits a near zero temperature glide at any composition, behaving as a pure fluid. Consequently, R513A becomes an excellent alternative to pure R134a. The addition of R1234yf decreases the GWP by 56%, approximately, compared to the original refrigerant R134a. In addition, recent studies demonstrate that R513A presents higher mass flow rate and cooling capacity (higher COPs) than R134a when evaporation temperatures between -15 and 12.5 °C are used, although at the price of a higher power consumption.⁵

While the combination of HFC–HFO blends is becoming a promising alternative to reduce the GWP of HFC refrigerants keeping adequate thermophysical properties, a remaining unsolved concern is related to the current null recyclability of fluorinated refrigerants. Typically, refrigeration equipment has a certain life period. After this, the refrigerant blend is removed from the circuit and, in most occasions, sent to incineration due to their low added value. Indeed, the separation of the compounds forming a blend would be desirable in order to reuse them (or at least one of them, the one with the lowest GWP and higher cost, the HFO), but current separation technologies, such as cryogenic distillation, are energy and cost intensive.⁶ In the search for alternative methods, ionic liquids (ILs) have been proposed as new extractive agents for absorption separation. It is well-known that their extremely low vapor pressures, high thermal and chemical stabilities, and wide liquid temperature ranges favor their use as solvents. Moreover, the huge amount of possible cation–anion combinations allows the design of task-specific fluids for a particular process.

In the literature, several studies have pointed out that the use of fluorinated anions contributes to increase the solubility of F-refrigerants.^{7,8} Here, it is important to highlight the pioneering work of Shiflett and co-workers on the study of the solubility of HFCs in ILs.^{9–14} From these contributions, several patents on the utilization of ILs for separation and recycling of refrigerants and fluorinated monomers were published.^{15–21} However, one of the major difficulties arises from obtaining a good selectivity that allows a proper separation of two compounds with similar thermophysical properties and chemical structures. In addition, far fewer studies have been performed with HFOs and blends,^{22–24} and their interactions with ILs are less known. In this regard, the use of reliable computational tools provides a quick manner to screen a vast amount of solvents and offer qualitative responses. This type of tools can significantly reduce the quantity of experiments as it is possible to evaluate many different solvents (ILs in this case) based on few experimental measurements through the use of transferable molecular parameters. The use of molecular simulations gives great details on the interactions at the molecular level, as it has been shown in a series of papers done for refrigerants by Raabe,^{25–29} at the price of an intensive computer cost. On the contrary, molecular-based equations of state, with less level of

detail, still keep the relevant information on the molecules at a coarse-grain level, providing rapid and reliable results. Consequently, they can be used as a first screening tool for a new separation technology. This is the approach followed in a recently published study where soft-SAFT was used to study the separation of the compounds forming the refrigerant blend R410A by selected ILs.³⁰

In a previous work, we have successfully modeled the vapor–liquid equilibria of azeotropic mixtures containing R134a and R1234yf at different conditions.³¹ Using the robust models developed in that work, we go one step further and focus here on investigating, for the first time, the separation of the R134a + R1234yf system from the commercial mixture R513A using selected ILs that act like an *entrainer* in an extractive distillation process. This requires that the selected solvents present a satisfactory selectivity toward one of the components versus the other to properly separate them. Hence, a previous investigation concerning the solubility of R513A's pure components into each IL is required. Based on the availability of experimental data, three ILs sharing the cation 1,3-hexylimidazolium, $[C_6C_{1im}]^+$, and three different fluorinated anions, tetrafluoroborate ($[BF_4]^-$), hexafluorophosphate ($[PF_6]^-$), and bistrifluoromethylsulfonylimide ($[TF_2N]^-$), have been selected to explore the ideal solvent with the best extraction properties when interacting with R513A. The main outcomes of this study are presented and discussed in this contribution.

2. THEORY

2.1. Soft-SAFT. The soft-SAFT equation of state (EoS)³² is a well-known variant of the original SAFT approach,^{33,34} which differs in the manner the reference term is treated. Soft-SAFT uses a Lennard-Jones (LJ) monomer as a reference fluid, where the van der Waals dispersive interactions are already included in a single expression. As the equation has been extensively revised in the bibliography, only the basic details are retained here. The reader is referred to the original soft-SAFT contributions for a deeper analysis of the equation.^{32,35}

Soft-SAFT is written in terms of the Helmholtz energy in a series of microscopic contributions based on statistical mechanics concepts.

$$A^{\text{res}} = A - A^{\text{id}} = A^{\text{ref}} + A^{\text{chain}} + A^{\text{assoc}} \quad (1)$$

where the subscripts id, ref, chain, and assoc refer to the ideal, reference, chain, and association terms, respectively. In other words, each term considers a physical effect in the molecular and adds a certain “energy” to the system.

For associating fluids, the terms in eq 1 include a total of five molecular parameters defining the fluid based on its structure at a “coarse-grained” scale. In particular, the reference term has a characteristic segment diameter (σ_{ii}) and dispersive energy (ϵ_{ii}/k_B) describing the interactions between two LJ monomers. The expression was fitted to molecular simulation data by Johnson et al.³⁶ The extension to mixtures of this term requires the use of the van der Waals one-fluid theory, where the cross-interaction parameters are calculated with the modified Lorentz–Berthelot combining rules:

$$\sigma_{ij} = \eta_{ij} \left(\frac{\sigma_{ii} + \sigma_{jj}}{2} \right) \quad (2)$$

$$\epsilon_{ij} = \xi_{ij} \sqrt{\epsilon_{ii} \epsilon_{jj}} \quad (3)$$

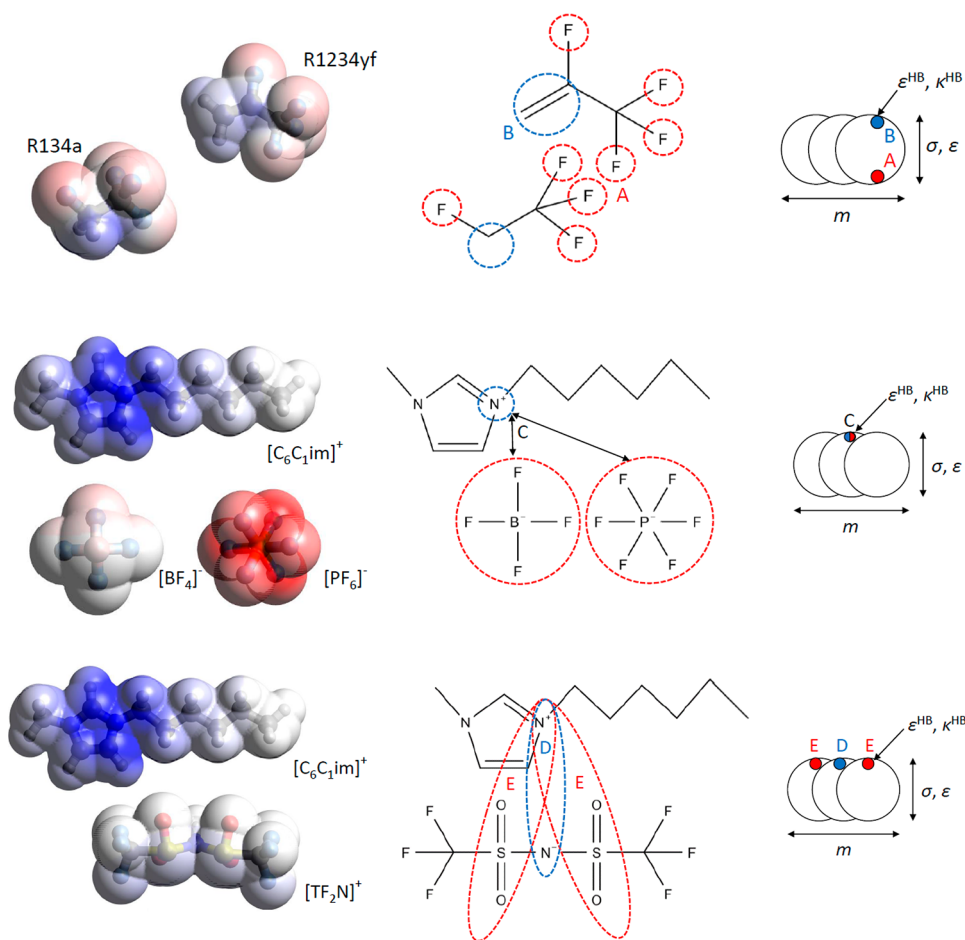


Figure 1. Atomistic representation coarse-grained identification of association sites and soft-SAFT molecular model of the different molecules studied in this work. (a) R134a and R1234yf, (b) [C₆C₁im][BF₄] and [C₆C₁im][PF₆], and (c) [C₆C₁im][TF₂N].

η_{ij} and ξ_{ij} (i.e., i and j are different species) are the size and energy binary parameters, respectively, which are typically fitted to binary mixture data. When η_{ij} and ξ_{ij} are equal to 1, the equation is used in a predictive manner for mixtures.

The chain and association terms are built from Wertheim's theory^{37–39} and can be expressed for mixtures (as a function of the composition) as

$$A^{\text{chain}} = RT \sum_i x_i (1 - m_i) \ln g_{ij}^{\text{LJ}} \quad (4)$$

$$A^{\text{assoc}} = RT \sum_i x_i \sum_{\alpha} \left(\ln X_{\alpha,i} - \frac{X_{\alpha,i}}{2} \right) + \frac{M_i}{2} \quad (5)$$

where T and R are the temperature and the ideal gas constant, respectively.

The chain term (eq 4) depends on the radial distribution function (g_{ij}^{LJ}) of the fluid and the chain length parameter (m_i) that describes the number of LJ spheres forming the chain. The association term (eq 5) depends on the total number of association sites of each component (M_i) and the fraction of molecules not bonded to the site α (X_{α}). This fraction can be solved through a mass action balance that depends on two additional parameters that define a site of association: the site–site bonding volume of association $K_{\alpha\beta,ii}^{\text{HB}}$ and the site–site association energy $\epsilon_{\alpha\beta,ii}^{\text{HB}}$. The cross-association interactions among association sites of different compounds are given by equivalent Lorentz–Berthelot combining rules.

$$K_{\alpha\beta,ij}^{\text{HB}} = \left(\frac{\sqrt[3]{K_{\alpha\beta,ii}^{\text{HB}}} + \sqrt[3]{K_{\alpha\beta,jj}^{\text{HB}}}}{2} \right)^3 \quad (6)$$

$$\epsilon_{\alpha\beta,ij}^{\text{HB}} = \sqrt{\epsilon_{\alpha\beta,ii}^{\text{HB}} \epsilon_{\alpha\beta,jj}^{\text{HB}}} \quad (7)$$

Although in some cases eq 7 is modified to include a binary parameter to modify the cross-association energy,^{31,40} this work has been performed using the predicted values obtained from these equations.

The accuracy of soft-SAFT has been tested for a variety of complex mixtures. In particular, it is interesting to highlight its application to reproduce the thermophysical properties of hydrofluorocarbons^{41,42} and hydrofluoroolefins³¹ as well as the solubility of greenhouse gases (CO₂, SO₂, N₂O, CO, HFCs) in many ILs.^{30,43–50}

2.2. Molecular Models. Following previous contributions,^{31,41} R134a and R1234yf are modeled as homonuclear chainlike molecules with two associating sites (A and B) of different natures simulating the dipole moment caused by the electronegativity (see Figure 1a for a graphical representation of the approximations done). Only A–B interactions are allowed. The same molecular model has been applied to HFC R152a, which has been included to highlight the transferability of the approach to similar systems (see section 3.1.2). The whole set of parameters for all compounds is taken from our

Table 1. Optimized Soft-SAFT Parameters for Selected HFCs, HFO, and ILs

	m	σ [Å]	ϵ/k_B [K]	ϵ^{HB}/k_B [K]	κ^{HB} [Å ³]	ref
R1234yf	1.136	4.808	173.4	1909	24050	31
R134a	1.392	4.166	166.6	1862	24050	31
R152a	1.392	4.007	170.9	1885	24050	31
[C ₆ C ₁ im][PF ₆]	5.095	4.210	423.0	3450	2250	47
[C ₆ C ₁ im][BF ₄]	5.005	4.110	423.0	3450	2250	47
[C ₆ C ₁ im][TF ₂ N]	6.338	4.334	404.2	3450	2250	51

previous work,³¹ and they are presented in Table 1 for completeness.

Following previous work,^{47,51,52,61} [BF₄][−] and [PF₆][−] imidazolium-based ILs are modeled as a single chain (based on the pair assumption principle) with a single associating site (named type C) mimicking cation–anion interactions (see Figure 1b).^{47,52} This site is of dual nature, so C–C type interactions are allowed. Finally, [TF₂N][−] imidazolium-based ILs are represented as a chain with three associating sites due to the delocalization of the anion electric charge.⁵¹ Consequently, one site (D) represents the nitrogen atom interactions with the cation, and two sites (E) account for the presence of the oxygen groups in the anion (see Figure 1c). Only D–E interactions are allowed. When ILs are mixed with HFC or HFO molecules, cross-associations between sites representing opposite (or dual nature) charges (A–C, B–C, A–D and B–E) are considered. The final sets of soft-SAFT parameters for the ILs used in this work, [C₆C₁im][PF₆], [C₆C₁im][BF₄], and [C₆C₁im][TF₂N], are taken from previous references^{47,51} and are also included in Table 1.

3. RESULTS

3.1. Binary Systems. **3.1.1. R513A.** The models and parameters for the two components (R134a and R1234yf) integrating R513A have been taken from our previous work³¹ in a transferable manner. Results concerning the vapor–liquid equilibrium (VLE) of the mixture are depicted in Figure 2 and compared to experimental data.⁵³ As the composition is constant, the VLE diagram looks like the one of a pure fluid. Figure 2a represents the liquid (right side) and vapor (left side) saturated densities as a function of the temperature. Both lines join at the critical point of the mixture, which corresponds to an exact point in the critical line of the system R134a + R1234yf. Figure 2b represents the vapor pressure of the mixture corresponding to each temperature up to the critical point. Soft-SAFT provides excellent agreement with experimental data for the whole range of temperatures with the use of two binary parameters (full line, $\xi = 0.960$ and $\eta = 0.985$). It is important to note that the VLE of the binary system R134a + R1234yf had already been studied in our previous work for the whole range of composition³¹ using only the energy binary parameter ($\xi = 0.960$); however, we have observed that the behavior of R513A is better reproduced when adding the size η binary parameter, correcting minor deviations in the saturated density of the mixture (see the dashed line description without adding η). The VLE of the binary system R134a + R1234yf at different temperatures with the two binary parameters used in this work is shown in Figure S1 of the Supporting Information, in excellent agreement with experimental data for the whole composition range.⁵⁴ The use of these binary parameters corrects the differences in molecular energy and volume of interactions between the two refrigerants in the mixture in an implicit manner; their values can be attributed to the different

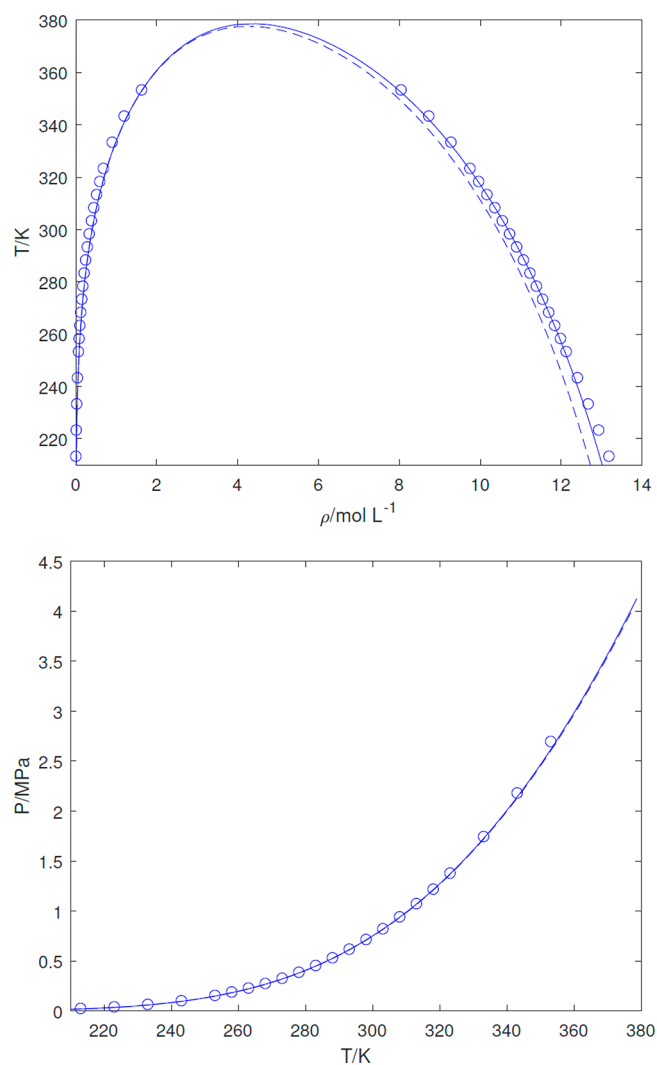


Figure 2. Phase behavior of R513A mixture (56 wt % R1234yf and 44 wt % R134a). (a) Vapor and liquid saturated density vs temperature. (b) Vapor pressure vs temperature. Symbols represent correlations to experimental data,⁵³ while lines correspond to the soft-SAFT modeling. The solid line is done with $\xi = 0.96$ and $\eta = 0.985$, while the dashed line is only using $\xi = 0.96$ ($\eta = 1$).

molecular interactions, especially those concerning polar interactions,⁵⁵ which in this work have been modeled with association.

3.1.2. Binary Systems R1234yf + [C₆C₁im][X] and R134a + [C₆C₁im][X]. In order to study the separation of R134a from R1234yf using ILs, the next step consists of the description of the solubility of them in the selected ILs. The molecular models and parameters of the pure components were taken from previous works in a transferable manner (see Table 1). All binary combinations between either R1234yf or R134a and

$[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$, $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$, or $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$ have been considered and compared to available experimental data. Although soft-SAFT can qualitatively describe the behavior of the mixtures, it is important to achieve quantitative agreement with the available data of the binary systems to properly assess the validity of these ILs for separation purposes. Consequently, one or two binary and temperature independent parameters η and ξ were used (see Table 2) for that purpose. Some relevant

Table 2. Binary Parameters ξ and η for the Binary Mixtures Studied in This Work and Reference of the Experimental Data Used for Fitting Them

blend	ξ	η	ref exptl data
R1234yf + R134a	0.960	0.985	54
R1234yf + $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$	1.160	1.000	23
R1234yf + $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$	1.170	1.000	22
R1234yf + $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$	1.140	1.000	24
R134a + $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$	1.215	1.040	6
R134a + $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$	1.225	1.040	6
R134a + $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$	1.160	1.040	6

trends can be identified in the values of these parameters and how this is related to the interactions between the different compounds in the mixture. For instance, a value of the energy binary parameter ξ greater than 1 is found in all cases, suggesting that the solubility of the refrigerant in the IL is higher than just the simple combination of the two energy parameters (when $\xi_{ij} = \xi$ in eq 3 would be equal to 1). Notwithstanding, those values appear to be in line with the ones attained in other publications for similar associative systems where, in all cases, the correction is mainly caused by the energetic contribution ξ .³⁰ As already mentioned in previous work,³⁰ this might be an indication that the model for the mixture is missing some interactions for this asymmetric system. However, it can be seen that the ξ value is very similar for the $[\text{BF}_4]^-$ and $[\text{PF}_6]^-$ -based ILs, which means that the parameter can be transferred from one anion to the other one in case no experimental data are available. The ξ value with the $[\text{TF}_2\text{N}]$ -based ILs is slightly lower, possibly because of the higher number of association sites used in the IL model. Concerning the size binary parameter, η , it is only used to correct the mixtures with R134a (no further fitting is necessary when using R1234yf) and the value is constant for all ILs, being close to unity.

As an example, results for $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ with R1234yf and R134a are plotted in Figures 3 and 4, respectively. The R1234yf + $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ system shows excellent agreement with the available experimental data²³ at all temperatures using only a constant $\xi = 1.160$ binary parameter (see Table 2), which is fitted to an intermediate isotherm at 303.15 K and used to predict the rest. The solubility of R1234yf in $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$ and $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$ is provided in Figures S2 and S3 in the Supporting Information.

Concerning the results of R134a + $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$, plotted in Figure 4, excellent agreement is attained by fitting constant binary parameters ξ and η (see Table 2) to 323.15 K, and predicting the other two isotherms of 298.15 and 348.15 K. As expected according to the experimental information,⁶ the solubility of this compound is higher than that of R1234yf. The solubility of R134a with the other nonselected ILs, $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$ and $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$, is also available in Figures S4 and S5 of the Supporting Information, respectively. In all cases,

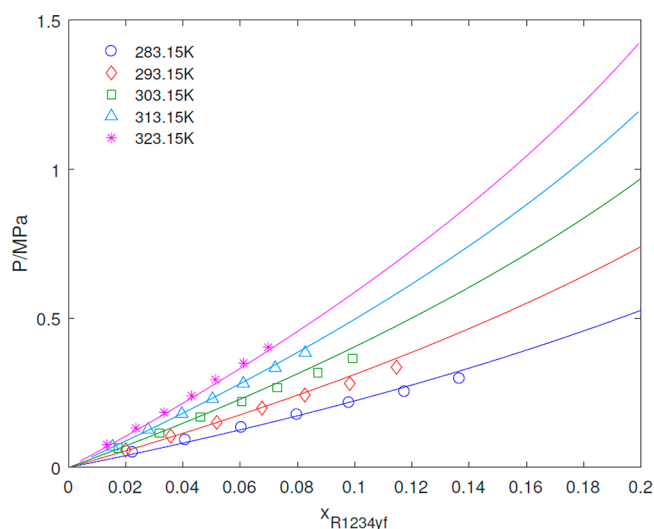


Figure 3. Pressure–composition diagram of the blend R1234yf (1) + $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ (2) at 283.15 (blue circle), 293.15 (red diamond), 303.15 (green square), 313.15 (light blue triangle), and 323.15 K (magenta asterisk). Symbols are experimental data from ref 23, and lines are soft-SAFT calculations.

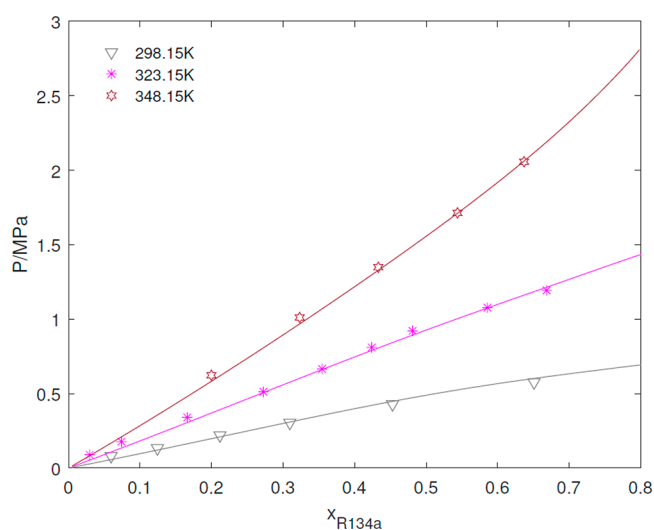


Figure 4. Pressure–composition diagram of the blend R134a (1) + $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ (2) at 298.15 (gray down triangle), 323.15 (magenta asterisk), and 348.15 K (deep red star). Symbols are experimental data from ref 6, and lines soft-SAFT calculations.

quantitative agreement is achieved between the model and the experimental data.⁶

Apart from the agreement among the experimental data and the soft-SAFT calculations, it is important to ensure the transferability and suitability of the approach proposed before proceeding to evaluate further thermodynamic properties and predict the behavior of ternary systems from the theoretical model. As an additional test to validate the performance of the equation, the prediction of the system R152a + $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$ from R134a + $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$ is provided. R152a has the same number of carbons and a similar structure, so it is believed that its mixture behavior in ILs should be similar to that of R134a. For this reason, the binary parameters are transferred from one mixture to the other one. The molecular parameters of pure R152a are taken from a previous contribution and are included in Table 1.³¹ A comparison

for both mixtures at two different temperatures is given in Figure 5. As it can be seen, there is perfect agreement between

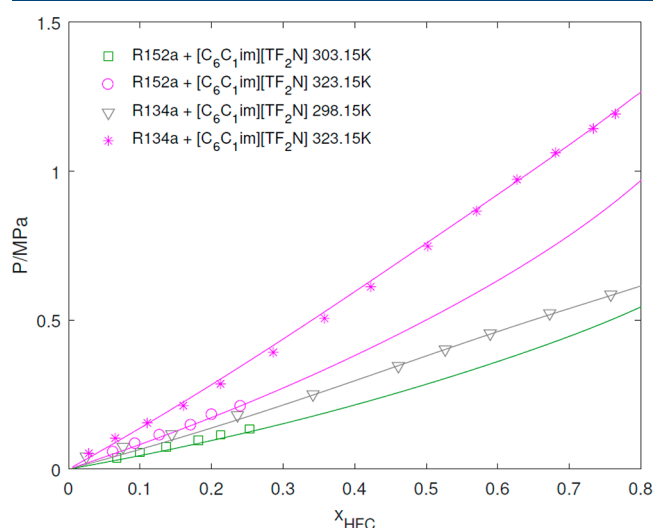


Figure 5. Pressure–composition diagram of R152a + [C₆C₁im]–[TF₂N] at 303.15 (green square) and 323.15 K (pink circle) and R134a + [C₆C₁im]–[TF₂N] at 298.15 (gray down triangle) and 323.15 K (pink asterisk).

the experimental data⁵⁶ and the soft-SAFT predictions to describe the solubility of R152a, without a deterioration of the performance of the equation when compared to the description of the R134a solubility.

3.2. Selectivity and Absorption Calculations. Once the VLEs of the six binary systems R1234yf + [C₆C₁im][BF₄], R1234yf + [C₆C₁im][PF₆], R1234yf + [C₆C₁im][TF₂N], R134a + [C₆C₁im][BF₄], R134a + [C₆C₁im][PF₆], and R134a + [C₆C₁im][TF₂N] have been reproduced in quantitative agreement with the experimental data, and the model has been validated, the selectivity of R134a and R1234yf in the correspondent ionic liquid is evaluated from these calculations, in order to decide which is the best solvent to carry out the separation of the R513A blend. Here, it is important to note that this is an ideal selectivity, as the calculation does not include the competition between the two compounds. Consequently, this can only be used for comparative purposes. The ideal selectivity at infinite dilution, α , relates the amount of refrigerant gas that is dissolved in the IL at a constant temperature and is expressed as the ratio of Henry's law constants between the two pure refrigerants.⁵⁷ Subsequently, soft-SAFT EoS is applied to predict pressure data as solubility approaches zero for each isotherm and to compute the ideal selectivity for the particular blend in terms of volatility or Henry's law constants:

$$\alpha_{R134a/R1234yf}^T = \frac{H_{R1234yf}^T}{H_{R134a}^T} = \frac{\lim_{x_{R1234yf} \rightarrow 0} \left(\frac{P_i}{x_{R1234yf}} \right)^T}{\lim_{x_{R134a} \rightarrow 0} \left(\frac{P_i}{x_{R134a}} \right)^T} \quad (8)$$

where T and P correspond to the absolute temperature and pressure, respectively, H_i is Henry's law constant referred to a compound i , x_i is the composition of compound i , and finally $\alpha_{i/j}$ reflects the ideal selectivity at infinite dilution of compound j with respect to i .

The results obtained for the ideal selectivity are plotted in Figure 5, for a temperature of 323.15 K, and confirm that R134a has a higher capacity of absorption than R1234yf. The highest selectivity, $\alpha_{R134a/R1234yf}$ is obtained with [C₆C₁im]–[PF₆] and approaches a value of 3. The Henry constants for both compounds are also included in Figure 5, showing a slightly higher solubility of both compounds in [C₆C₁im]–[BF₄], although its selectivity remains lower than that of [C₆C₁im]–[PF₆]. The performance of [C₆C₁im]–[TF₂N] is worse than those of the other two ILs in terms of both absorption and selectivity.

Other useful thermodynamic tools to illustrate the absorption of R513A into the ILs are the absorption enthalpy, $\Delta_{\text{dis}}H$, and entropy, $\Delta_{\text{dis}}S$, which provide information on the strength of the interactions and the degree of order when a gas is dissolved in the solvent, respectively.⁵⁸ A negative solvation enthalpy and a positive solvation entropy will favor the dissolution of the refrigerant in the IL. These variables can be calculated using eqs 9 and 10.

$$\Delta_{\text{dis}}H = \bar{H} - H^{\text{ig}} = R \left| \frac{\partial \ln P}{\partial 1/T} \right|_{x_1} \sim R \left| \frac{\Delta \ln P}{\Delta 1/T} \right|_{x_1} \quad (9)$$

$$\Delta_{\text{dis}}S = \bar{S} - S^{\text{ig}} = -R \left| \frac{\partial \ln P}{\partial \ln T} \right|_{x_1} \sim -R \left| \frac{\Delta \ln P}{\Delta \ln T} \right|_{x_1} \quad (10)$$

where $\bar{H}$, $\bar{S}$, H^{ig} , and S^{ig} are the partial molar enthalpy and entropy of refrigerant gas in solution and the enthalpy and entropy of the pure gas in the ideal gas phase, P is the gas partial pressure of the component, T is the overall system temperature, and x_1 is the mole fraction of the gas dissolved in the IL, which is fixed at 0.01 (a low value) to use the approximated expressions.

The results for all binary mixtures using soft-SAFT and eqs 9 and 10 are provided in Table 3. R134a exhibits higher negative values of both the molar enthalpy and entropy of solvation than R1234yf in all cases. The values for the molar enthalpy range between –20.55 and –24.04 kJ/mol for R134a and between –16.27 and –20.82 kJ/mol for R1234yf. These values fall within the range of physical absorption although they are higher than the values obtained for other gases, such as CO₂, in these ILs.⁵⁹ The lowest molar enthalpy of solvation is found for

Table 3. Calculated Enthalpy and Entropy of Dissolution for R1234yf and R134a in [C₆C₁im][PF₆], [C₆C₁im][BF₄], and [C₆C₁im][TF₂N] from 283.15 to 348.15 K at an IL Liquid Phase Molar Composition of 0.01

IL	$\Delta_{\text{dis}}H/\text{kJ}\cdot\text{mol}^{-1}$			$\Delta_{\text{dis}}S/\text{J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$	
	R134a	R1234yf	$\Delta_{\text{dis}}H^{\text{HFC}} - \Delta_{\text{dis}}H^{\text{HFO}}$	R134a	R1234yf
[C ₆ C ₁ im][PF ₆]	–21.03	–16.27	–4.76	–68.25	–52.81
[C ₆ C ₁ im][BF ₄]	–20.55	–16.16	–4.39	–66.69	–52.44
[C ₆ C ₁ im][TF ₂ N]	–24.04	–20.82	–3.22	–78.01	–67.56

$[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$, followed by $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ and $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$. This can be due to the delocalization of the negative charge in the anion caused by the electronegativity of the oxygen and fluorine atoms, causing the withdrawal of the IL anion from the cation, which would allow a better enthalpic interaction with R134a and R1234yf. However, for gas separation purposes, it is interesting to check the difference of molar enthalpy of both substances, also given in Table 3. It is observed that, in this case, $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ shows the major difference, suggesting that this IL would facilitate the separation in the sense that it would favor the absorption of R134a over R1234yf to a higher extent. This agrees with the higher ideal selectivity found in Figure 6.

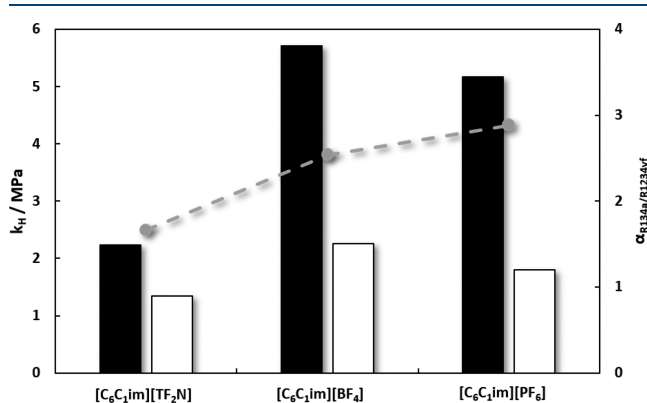


Figure 6. Calculated Henry's constants for R1234yf (black) and R134a (white) and ideal selectivities (gray) in $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$, $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$, and $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$ at 323.15 K.

Concerning the molar entropy of solvation, negative values are found in all cases as a result of the condensation of the gas. The reduction of entropy due to this phase change is not compensated by the entropy generated as a consequence of the disruption of the refrigerant into the ordered structure of the IL. Moreover, considering that the three studied ILs have the same alkyl chain length, the discussion is reduced to the difference caused by the anion. Interestingly, the molar entropy is lower for $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$ than for the other two ILs. Possibly, the size of this anion hinders an ordered arrangement, generating free spaces that are occupied by the gases, while in the cases of $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ and $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$, which have compact anions, the solubilization of the gas generates a higher degree of entropy.

On the basis of the obtained results, the $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ IL has been preselected to evaluate the competitive selectivity between both refrigerants by the predictive characterization of the ternary blend among R513A and $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$, as it exhibits the highest ideal selectivity and the highest difference of molar enthalpy between both compounds.

3.3. Ternary System R513A + $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$. Once $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ has been chosen as the entrainer to separate R513A into its pure components, and the corresponding binary VLE characterizations have been done, the prediction of the ternary VLE is carried out in order to evaluate the competitive selectivity between both compounds at the desired ratio. Figure 7 displays the ternary phase diagram at two temperatures, 298.15 and 323.15 K, and different pressures. While the diagram predicts the equilibrium liquid phase composition at any proportion among R1234yf, R134a, and $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$, the reader is referred to the dashed black line

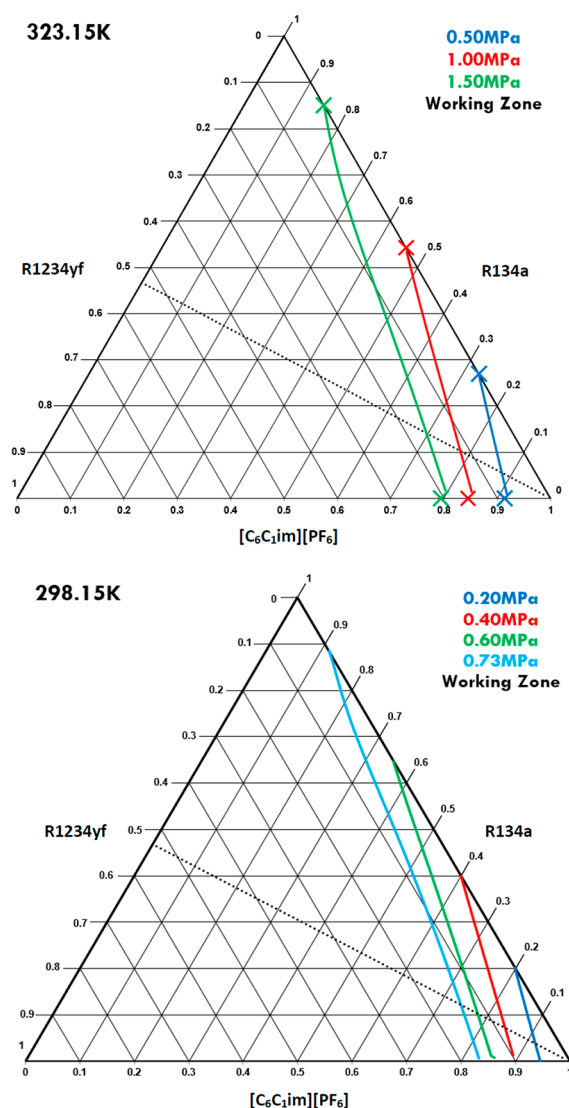


Figure 7. Ternary diagram of the mixture R1234yf, R134a, and $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ at (a) 323.15 K and 0.50 (blue), 1.00 (red), and 1.50 MPa (green). Predicted liquid phase (cross) data from the binaries (this work). Ternary diagram at (b) 298.15 K and 0.20 (blue), 0.40 (red), 0.60 (green), and 0.73 MPa (light blue). Degraded black line refers to the R513A composition region.

to evaluate the saturation properties for the commercial mixture R513A (56 wt % R1234yf, 44 wt % R134a). Different isobars have been plotted in order to check the effect of the pressure in the phase equilibria. The competitive selectivity between both gases can be evaluated from the results depicted in the ternary diagram.⁶⁰ The expression, written in terms of the selectivity of R134a with respect to R1234yf, reads

$$S_{\text{R134a/R1234yf}} = \frac{y_{\text{R1234yf}}/x_{\text{R1234yf}}}{y_{\text{R134a}}/x_{\text{R134a}}} \quad (11)$$

where x and y are the mole fractions of HFC or HFO in the liquid or vapor phase, respectively.

The results obtained for the selectivity at different IL feed compositions (z_{IL}) of an R513A mixture, represented by the symbols, are plotted in Figure 8. First, it is interesting to note that the competitive selectivity and the ideal one have practically the same values (compared at 323.15 K), so the

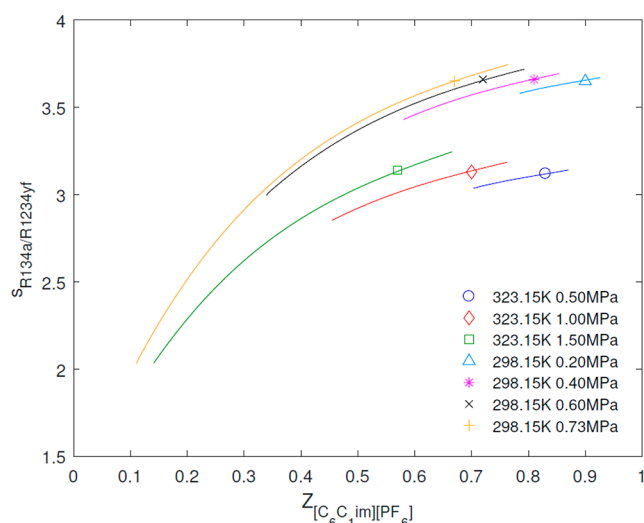


Figure 8. Competitive selectivity as a function of ionic liquid feed composition at 323.15 K and 0.50 (blue circle), 1.00 (red diamond), and 1.50 MPa (green square) and at 298.15 K and 0.20 (light blue triangle), 0.40 (magenta asterisk), 0.60 (black cross), and 0.73 MPa (orange plus). Symbols represent R513A composition at each pressure and temperature.

presence of the second gas does not seem to have a strong influence in the absorption capacity. In Figure 8, all values fall into two different levels, 3.15 and 3.65 approximately, influenced by the temperature. The selectivity increases at lower temperatures with an average improvement of 17.3% from 323.15 K until 298.15 K. The influence of pressure seems to be negligible in terms of selectivity, but at the cost of incrementing the amount of ionic liquid in the system. Consequently, it is recommended to work at low temperature and moderate pressure conditions to carry out this separation with $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$.

4. CONCLUSIONS

In this contribution, an in-depth thermodynamic characterization of the absorption capability of three ionic liquids as possible solvents for an efficient separation of R513A compounds has been carried out using the soft-SAFT equation of state. In particular, the solubilities of R134a and R1234yf, compounds forming the R513A mixture, in the fluorinated anions $[\text{BF}_4]^-$, $[\text{PF}_6]^-$, and $[\text{TF}_2\text{N}]^-$ sharing the common cation $[\text{C}_6\text{C}_1\text{im}]$, have been accurately described by the soft-SAFT model, reproducing the available experimental data and validating the model for the ternary systems. Cross-association interactions between the refrigerant gases and the ILs were explicitly considered. In addition, a temperature independent energy binary parameter was fitted to improve the calculations. A second size binary parameter, constant for all ILs, was added to better describe the solubility of R134a. The model is very accurate at different temperatures and pressures. Its robustness was checked by predicting the solubility of R152a in an IL transferring the binary parameters from R134a. Once the model was validated, it was used to evaluate the Henry constants, the ideal selectivity, and the enthalpy and entropy of dissolution of the gases in the three selected ILs. Results revealed that $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ is the most promising IL to perform the desired separation. Hence, the ternary mixture between both gases and $[\text{C}_6\text{C}_1\text{im}][\text{PF}_6]$ was predicted in the absence of experimental data at two different temperatures to

assess the feasibility of the recovery of the two separated components by this IL. The influence of the temperature, pressure, and ionic liquid feed composition in the competitive selectivity was studied, concluding that the separation is favored at low temperature, while a balance between the pressure and the amount of IL must be checked to find an optimal condition of separation. This study is part of a complete characterization of alternative solvents for an efficient separation of azeotropic mixtures of blends of refrigerants, with the aim to reuse those fluids with a lower global warming potential. It shows how molecular modeling tools, such as soft-SAFT, can be used even in the absence of experimental data as a first screening method; then experiments can be carried out for the selected entrainer to further validate the approach and refine the optimal conditions for the separation.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.0c02331>.

Pressure–composition diagrams for binary systems R134a + R1234yf, R1234yf + $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$, R1234yf + $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$, R134a + $[\text{C}_6\text{C}_1\text{im}][\text{BF}_4]$, and R134a + $[\text{C}_6\text{C}_1\text{im}][\text{TF}_2\text{N}]$ (PDF)

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Notes

The authors declare no competing financial interest.

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